1 Conformal Changes of Riemannian Metrics

We assume that the reader is familiar with the basic notions of smooth manifolds, Riemannian manifolds, Riemannian metrics, and Levi–Civita connections.

1.1 Introduction

We study the transformation laws for geometric quantities under a conformal change of metric. Let (M^n, g) be an n-dimensional Riemannian manifold, and consider the conformally related metric

$$\widetilde{g} = e^{2f} g,$$

where $f \in C^\infty(M)$.

The Levi–Civita connection, Riemann curvature tensor, Ricci curvature tensor, and scalar curvature associated with g will be denoted by

$$\nabla, \qquad \mathrm{Rm}, \qquad \mathrm{Ric}, \qquad \mathrm{Sc},$$

respectively. The corresponding quantities with respect to $\widetilde{g}$ will be denoted by

$$\widetilde{\nabla}, \qquad \widetilde{\mathrm{Rm}}, \qquad \widetilde{\mathrm{Ric}}, \qquad \widetilde{\mathrm{Sc}}.$$

1.2 Sign Conventions

Our convention for the Riemann curvature tensor is

$$\mathrm{Rm}(X, Y)Z = \nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z + \nabla_{[X,Y]} Z.$$

The Ricci curvature tensor is defined by

$$\mathrm{Ric}(X, Y) = \sum_{i=1}^{n} g\big(\mathrm{Rm}(X, e_i)Y, e_i\big),$$

where $\{e_i\}_{i=1}^n$ is a local g-orthonormal frame.

The scalar curvature is defined by

$$\mathrm{Sc} = \operatorname{tr}_g \mathrm{Ric} = \sum_{i=1}^{n} \mathrm{Ric}(e_i, e_i).$$

Equivalently,

$$\mathrm{Sc} = \sum_{i,j=1}^{n} g\big(\mathrm{Rm}(e_i, e_j)e_i, e_j\big).$$

With these conventions, a Riemannian manifold of constant sectional curvature K satisfies

$$\mathrm{Ric} = (n-1)Kg, \qquad \mathrm{Sc} = n(n-1)K.$$

For a smooth function f, its Hessian is defined by

$$\nabla^2 f(X, Y) = g(\nabla_X \nabla f, Y),$$

and the Laplace–Beltrami operator is defined by

$$\Delta f = \operatorname{tr}_g(\nabla^2 f).$$

1.3 Hypersurface Conventions

Let $\Sigma^{n-1} \subset M^n$ be an immersed hypersurface. With respect to g, denote its unit normal, second fundamental form, and mean curvature by

$$\nu, \qquad \mathrm{I\!I}, \qquad H,$$

respectively.

We adopt the convention

$$\mathrm{I\!I}(X,Y) = g(\nabla_X \nu, Y), \qquad X, Y \in T\Sigma,$$

and define the mean curvature by

$$H = \operatorname{tr}_{g_\Sigma} \mathrm{I\!I},$$

where g_Σ is the metric induced on Σ by g. Thus, H is the trace of the second fundamental form, rather than its normalized average.

The corresponding quantities computed with respect to $\widetilde{g}$ will be denoted by

$$\widetilde{\nu}, \qquad \widetilde{\mathrm{I\!I}}, \qquad \widetilde{H}.$$

Our goal is to derive the transformation laws for these intrinsic and extrinsic geometric quantities under the conformal change

$$g \longmapsto \widetilde{g} = e^{2f} g.$$

1.4 Conformal Transformation Formulas

Definition 1.1 (Kulkarni–Nomizu product)**.** *Let P and Q be symmetric $(0,2)$-tensors on M. Their Kulkarni–Nomizu product is the $(0,4)$-tensor $P \owedge Q$ defined by*

$$\begin{aligned}(P \owedge Q)(X,Y,Z,W) &= P(X,Z)Q(Y,W) + P(Y,W)Q(X,Z) \\ &\quad - P(X,W)Q(Y,Z) - P(Y,Z)Q(X,W).\end{aligned}$$

We regard the Riemann curvature tensor as a $(0,4)$-tensor by setting

$$\mathrm{Rm}(X,Y,Z,W) = g\big(\mathrm{Rm}(X,Y)Z, W\big).$$

Proposition 1.2 (Conformal transformation formulas)**.** *Let*

$$\widetilde{g} = e^{2f} g, \qquad f \in C^\infty(M).$$

Then, for any smooth vector fields X, Y on M, the Levi–Civita connections of g and $\widetilde{g}$ satisfy

$$\widetilde{\nabla}_X Y = \nabla_X Y + X(f)Y + Y(f)X - g(X,Y)\nabla f.$$

Regarding the Riemann curvature tensors as $(0,4)$-tensors, one has

$$\boxed{\widetilde{\mathrm{Rm}} = e^{2f}\left[\mathrm{Rm} - g \owedge \left(\nabla^2 f - df \otimes df + \frac{1}{2}|\nabla f|_g^2\, g\right)\right].}$$

Equivalently, for any smooth vector fields X, Y, Z, W on M,

$$\begin{aligned}\widetilde{\mathrm{Rm}}(X,Y,Z,W) = e^{2f}\Big\{ &\mathrm{Rm}(X,Y,Z,W) \\ &- g(X,Z)\nabla^2 f(Y,W) - g(Y,W)\nabla^2 f(X,Z) \\ &+ g(X,W)\nabla^2 f(Y,Z) + g(Y,Z)\nabla^2 f(X,W) \\ &+ g(X,Z)Y(f)W(f) + g(Y,W)X(f)Z(f) \\ &- g(X,W)Y(f)Z(f) - g(Y,Z)X(f)W(f) \\ &- |\nabla f|_g^2\big(g(X,Z)g(Y,W) - g(X,W)g(Y,Z)\big)\Big\}.\end{aligned}$$

The Ricci curvature tensors satisfy

$$\boxed{\widetilde{\mathrm{Ric}} = \mathrm{Ric} - (n-2)\left(\nabla^2 f - df \otimes df + \frac{1}{2}|\nabla f|_g^2\, g\right) - \left(\Delta f + \frac{n-2}{2}|\nabla f|_g^2\right) g.}$$

The scalar curvatures satisfy

$$\boxed{\widetilde{\mathrm{Sc}} = e^{-2f}\left[\mathrm{Sc} - 2(n-1)\left(\Delta f + \frac{n-2}{2}|\nabla f|_g^2\right)\right].}$$

Let $\Sigma^{n-1} \subset M^n$ be an immersed hypersurface. Then

$$\boxed{\widetilde{\nu} = e^{-f}\nu.}$$

For all $X, Y \in T\Sigma$, the second fundamental forms satisfy

$$\boxed{\widetilde{\mathrm{II}}(X,Y) = e^{f}\left[\mathrm{II}(X,Y) + \nu(f) g(X,Y)\right].}$$

Finally, the mean curvatures satisfy

$$\boxed{\widetilde{H} = e^{-f}\left[H + (n-1)\nu(f)\right].}$$